



Montage Wi-Fi M88WI6700 SDK Driver Compilation Guide for Linux OS ver. 1.0.0

Release History

Version	Date	Remarks
1.0.0	2017/02/16	Initial Release

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1. Overview

Montage m88wi6700 has two model : m88wi6700u with USB interface and m88wi6700s with SDIO interface.

This driver only supports the Linux Kernel version 3.13 and 3.14. The backports system can be used to support the m88wi6700 driver in the older version kernel (ex: 2.6.32). But the m88wi6700 driver doesn't support the kernel which version is newer than 3.14.

2. SDK Files

The SDK file name format is **m88wi6700_linux_driver_<YYYYMMDD>.tar.xz**.

Ex: m88wi6700_linux_driver_20170215.tar.xz

The files in the SDK is shown as follows:

```
m88wi6700_linux_sdk_YYYYMMDD.tar.xz
├── linux ──┬── backports-3.13
│           ├── host
│           └── util ── mont_ctl
```

The **host** folder is the M88WI6700 main driver. And the **backports-3.13** folder is the backport system to support the platform with the kernel order than 3.13.

The **util/mont_ctl** folder is the M88WI6700 private tools to do IO control.

3. Build the Driver without Backports system

3.1 Note

The platform which kernel version matched 3.13 or 3.14 can directly use the driver without backports system.

3.2 Config Options

3.2.1 Path

./host/Kconfig.rule

3.2.2 Configuration Definition

(1) CONFIG_LYNX_A1

Set the m88wi6700 firmware as A1 version.

- (2) CONFIG_LYNX_IBSS
Enable the IBSS feature.
- (3) CONFIG_LYNX_DEBUG
Enable the debug function.
- (4) CONFIG_LYNX_W2USB
Set the driver to be configured as m88wi6700u (USB).
- (5) CONFIG_LYNX_W2USB_FIRMWARE_ARRAY
Set the firmware source form the header file. Depend on CONFIG_LYNX_W2USB.
- (6) CONFIG_LYNX_W2SDIO
Set the driver to be configured as m88wi6700s (SDIO).
- (7) CONFIG_LYNX_W2SDIO_FIRMWARE_ARRAY
Set the firmware source form the header file. Depend on CONFIG_LYNX_W2SDIO.
- (8) CONFIG_LYNX_OS_LINUX
Set the platform OS as the Linux.
- (9) CONFIG_LYNX_WM_MANAGER
Use the wifi manager to control WiFi opertaion. Only be enabled in the real-time OS.
- (10) CONFIG_WPA
Enable the WPA feature. Depend on CONFIG_LYNX_WM_MANAGER.
- (11) CONFIG_HOST_WPS
Enable the WPS feature.
- (12) CONFIG_HOST_P2P
Enable the WiFi Direct feature.
- (13) CONFIG_SUPPORT_MONITOR_MODE
Enable the WiFi monitor mode.
- (14) CONFIG_LYNX_POWER_SAVING
Set the power save mode in the driver. Default is controlled by the device firmware.
- (15) CONFIG_SDIO_POLLING
Use the polling method in the data rx path.
- (16) CONFIG_TIMER_HANDLER
Enable the timer task to handle some exception case.

3.2.3 Example for m88wi6700u (USB interface)

```
CONFIG_LYNX_A1=y
CONFIG_LYNX_IBSS=y
CONFIG_LYNX_DEBUG=n
CONFIG_LYNX_W2USB_FIRMWARE_ARRAY=y
CONFIG_LYNX_W2USB=m
CONFIG_LYNX_W2SDIO_FIRMWARE_ARRAY=n
CONFIG_LYNX_W2SDIO=n
CONFIG_LYNX_OS_LINUX=y
CONFIG_LYNX_WM_MANAGER=n
CONFIG_WPA=n
CONFIG_HOST_WPS=n
CONFIG_HOST_P2P=n
CONFIG_SUPPORT_MONITOR_MODE=n
CONFIG_LYNX_POWER_SAVING=n
CONFIG_SDIO_POLLING=n
CONFIG_TIMER_HANDLER=y
```

3.2.4 Example for m88wi6700s (SDIO interface)

```
CONFIG_LYNX_A1=y
CONFIG_LYNX_IBSS=y
CONFIG_LYNX_DEBUG=n
CONFIG_LYNX_W2USB_FIRMWARE_ARRAY=n
CONFIG_LYNX_W2USB=n
CONFIG_LYNX_W2SDIO_FIRMWARE_ARRAY=y
CONFIG_LYNX_W2SDIO=m
CONFIG_LYNX_OS_LINUX=y
CONFIG_LYNX_WM_MANAGER=n
CONFIG_WPA=n
CONFIG_HOST_WPS=n
CONFIG_HOST_P2P=n
CONFIG_SUPPORT_MONITOR_MODE=n
CONFIG_LYNX_POWER_SAVING=n
CONFIG_SDIO_POLLING=y
CONFIG_TIMER_HANDLER=y
```

3.3 Make Driver and Generate the Kernel Module

After setup the Kconfig.rule, directly do **make** in the **host** folder. And the m88wi6700s.ko (for sdio) or m88wi6700u.ko (for usb) is generated, that is the driver kernel module.

Example:

```
host$ make
...
...
Building modules, stage 2.
MODPOST 1 modules
make[2]: Leaving directory `/usr/src/linux-headers-3.13.11.11'
#@make -C /lib/modules/3.13.11.11/build/ M= modules
make[1]: Leaving directory `<WORK FOLDER>/host'
host$
host$ ls m88wi6700s.ko
m88wi6700s.ko
```

4. Build the Driver with Backports system

4.1 Note

The backports system is needed when the developers use the M88WI6700 in the platform with the kernel which version is older than 3.13.

4.2 Makefile

The `./backports-3.13/Makefile` contains the **KLIB** and **KLIB_BUILD** info. The KLIB should be assigned to the project 's kernel path. And the KLIB_BUILD should be assigned to the project 's kernel's headers and config path.

KLIB & KLIB_BUILD in the Makefile:

```
ifneq ($(origin KLIB), undefined)
KMODPATH_ARG := "INSTALL_MOD_PATH=$(KLIB)"
else
KLIB := /lib/modules/$(shell uname -r)/
KMODPATH_ARG :=
endif
KLIB_BUILD ?= $(KLIB)/build/
```

Linux PC Environment:

```
KLIB := /lib/modules/3.13.0-51-generic/
KLIB_BUILD := /lib/modules/3.13.0-51-generic/build/
```

Example for the cross compiler project:

```
BACKPORT_PWD := $(shell pwd)
KLIB := $(BACKPORT_PWD)/../../linux2.6.29.4/
KLIB_BUILD ?= $(KLIB)/
```

The kernel referenced **MUST** been built before making the backports modules.

4.3 Config Options

4.3.1 Config File

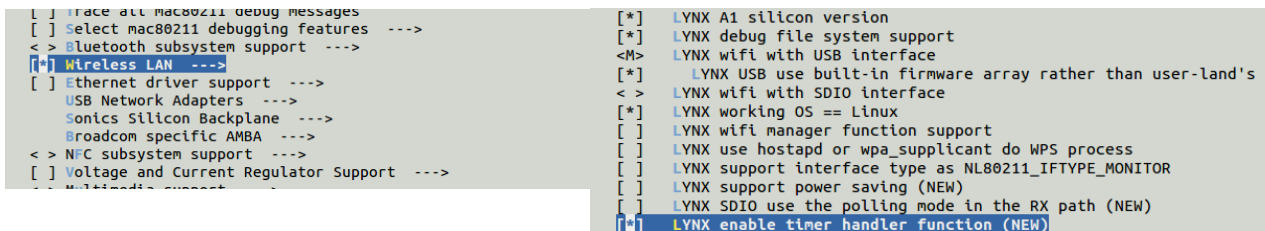
The default configure for backports is in the `./host/Kconfig`. The configuration meaning can be reference form *3.2.2 Configuration Definition* section.

4.3.2 Modify the Config Option Setting

Method 1:

In the backport folder, do **make menuconfig**. The options' path is in the "Wireless LAN".
Execution Example:

```
backports-3.13$ make menuconfig
make[2]: `mconf' is up to date.
```



```
[ ] Trace all mac80211 debug messages
[ ] Select mac80211 debugging features --->
< > Bluetooth subsystem support --->
[*] Wireless LAN --->
[ ] Ethernet driver support --->
  USB Network Adapters --->
    Sonics Silicon Backplane --->
    Broadcom specific AMBA --->
< > NFC subsystem support --->
[ ] Voltage and Current Regulator Support --->
- Multimedia support --->
[*] LYNX A1 silicon version
[*] LYNX debug file system support
<M> LYNX wifi with USB interface
[*] LYNX USB use built-in firmware array rather than user-land's
< > LYNX wifi with SDIO interface
[*] LYNX working OS == Linux
[ ] LYNX wifi manager function support
[ ] LYNX use hostapd or wpa_supplicant do WPS process
[ ] LYNX support interface type as NL80211_IFTYPE_MONITOR
[ ] LYNX support power saving (NEW)
[ ] LYNX SDIO use the polling mode in the RX path (NEW)
[*] LYNX enable timer handler function (NEW)
```

Method 2:

Modify the `.config` file in the `./backports-3.13` folder.

The config option name is modify from `CONFIG_XXXX` to `CPTCFG_XXXX` in the backports.

```
CPTCFG_LYNX_A1=y
# CPTCFG_LYNX_DEBUG is not set
CPTCFG_LYNX_W2USB=m
CPTCFG_LYNX_W2USB_FIRMWARE_ARRAY=y
# CPTCFG_LYNX_W2SDIO is not set
CPTCFG_LYNX_OS_LINUX=y
# CPTCFG_LYNX_WM_MANAGER is not set
# CPTCFG_HOST_WPS is not set
# CPTCFG_SUPPORT_MONITOR_MODE is not set
# CPTCFG_LYNX_POWER_SAVING is not set
# CPTCFG_SDIO_POLLING is not set
CPTCFG_TIMER_HANDLER=y
```

4.4 Make Driver and Generate the Kernel Module

Directly do make in the backports folder.

```
backports-3.13$ make
```

And the main kernel modules are built :

```
./compat/compat.ko
./net/wireless/cfg80211.ko
./drivers/net/wireless/lynx/m88wi6700u.ko
```